

Protein Domain Analysis of Hemoglobin Alpha Chain

Description

This project involved analyzing the protein domains of the human hemoglobin alpha chain using the InterPro tool to identify its functional regions.

Method

- Used InterPro protein analysis platform
- Input human hemoglobin alpha protein sequence
- Examined protein families, domains, and conserved regions

Results

The analysis identified the protein as a member of the hemoglobin family, specifically Hb-alpha-like proteins.

A well-defined **globin domain** was detected across the sequence and confirmed by multiple databases.

Additionally, conserved residues were observed, indicating structurally and functionally important regions.

Screenshot

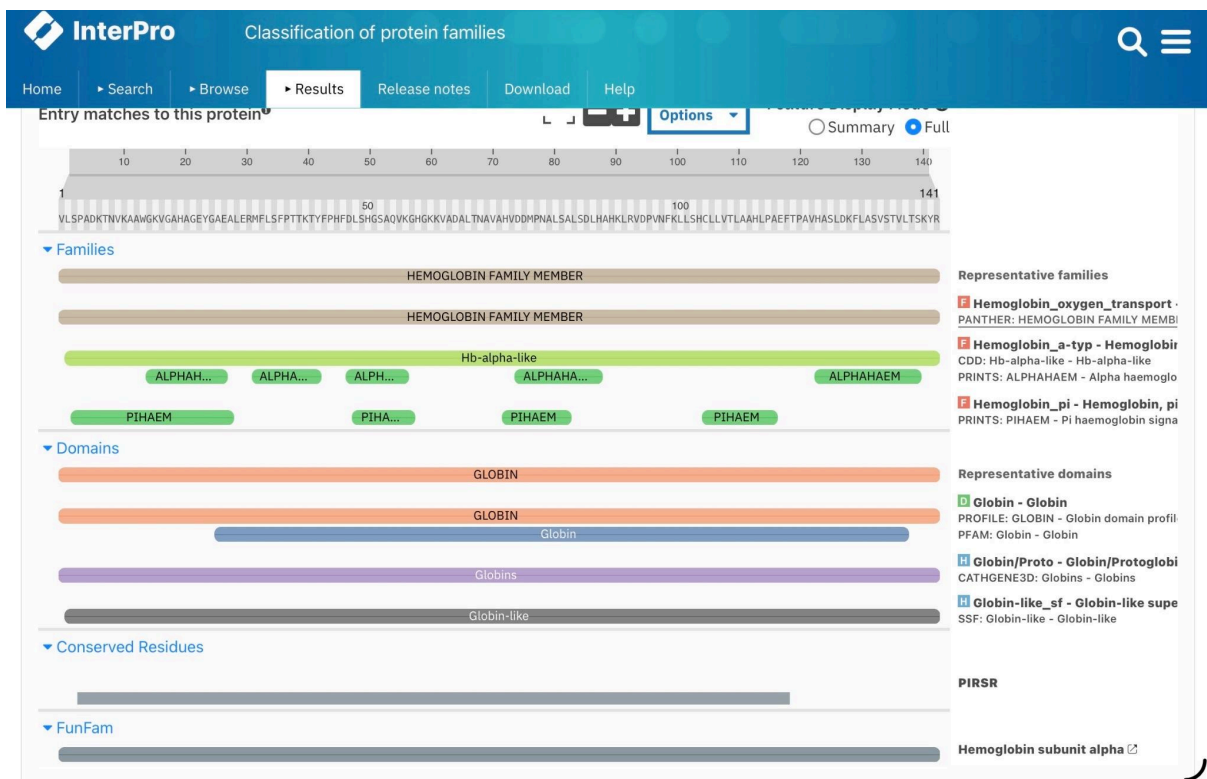


Figure 1: InterPro results showing the globin domain and conserved regions in human hemoglobin, confirming its role in oxygen binding and transport.

Interpretation

The presence of the globin domain confirms that the protein is involved in oxygen binding and transport.

The conserved regions further support its critical biological role, demonstrating how protein domain analysis can be used to infer function from sequence data.

Skills Demonstrated

- Protein domain analysis
- Functional prediction from sequence
- Interpretation of conserved regions
- Use of bioinformatics databases